[1]:
```
a=[10,20,30,40.5,"hello"]
for i in a:
    print(i)
```

```
10
20
30
40.5
hello
```

[2]:
```
a=["anu","banu","chitra"]
print(a)
for i in a:
    print(i)
```

```
['anu', 'banu', 'chitra']
anu
banu
chitra
```

[3]:
```
a=["anu","banu","chitra"]
print(a)
for i in range(len(a)):
    #print(i,a[i])
    print(f" index={i} , value={a[i]}")
```

```
['anu', 'banu', 'chitra']
 index=0 , value=anu
 index=1 , value=banu
 index=2 , value=chitra
```

[4]:
```
t=(10,20,30,40.5,"hello")
for i in t:
    print(i)
```

```
10
20
30
40.5
hello
```

[4]:
```
t=(10,20,30,40.5,"hello")
for i in t:
    print(i)
```

```
10
20
30
40.5
hello
```

[5]:
```
t=(10,20,30,40.5,"hello")
for i in range(len(t)):
    print(i,t[i])
```

```
0 10
1 20
2 30
3 40.5
4 hello
```

[6]:
```
s={10,20,30,40.5,"hello"}
for i in s:
    print(i)
```

```
20
40.5
10
hello
30
```

[7]:
```
d={'name':"anu","mark":60,'age':24}
print(d)
for key in d:
    print(key,d[key])
```

```
{'name': 'anu', 'mark': 60, 'age': 24}
name anu
mark 60
age 24
```

[8]:
```
d={'name':"anu","mark":60,'age':24}

for keys,value in d.items():
    print(keys, value)
```

```
name anu
mark 60
age 24
```

[9]:
```
a={"anu","hello",54,45.587}
for index,value in enumerate(a):
    print(index,value)
```

```
0 anu
1 hello
2 45.587
3 54
```

[10]:
```
a={00:"anu",10:"hello",20:54,30:45.587}
for index,(key,value) in enumerate(a.items()):
    print(index,(key,value))
```

```
0 (0, 'anu')
1 (10, 'hello')
2 (20, 54)
3 (30, 45.587)
```

[11]:
```
d1=[(1,2),(3,3,3),(4,4,4)]
print(d1)
d1.index((3,3,3))
# d1.index([3,3,3])
```

```
[(1, 2), (3, 3, 3), (4, 4, 4)]
```

[11]:
```
1
```

[18]:
```
a=[1,2]
b=(3,4,5)
c={6,7}
d={0:"zero",1:"one"}
merge_list=[a,b,c,d]
print(merge_list)
merge_tuple=(a,b,c,d)
```

[23]:
```
a=[1,2]
b=(3,4,5)
c={6,7}
d={0:"zero",1:"one"}
merge_list=[a,b,c,d]
print(merge_list)
merge_tuple=(a,b,c,d)
print(merge_list)
merge_dict={1:'a',2:'b',3:'c',4:'d'}
print(merge_list)
```

```
[[1, 2], (3, 4, 5), {6, 7}, {0: 'zero', 1: 'one'}]
[[1, 2], (3, 4, 5), {6, 7}, {0: 'zero', 1: 'one'}]
[[1, 2], (3, 4, 5), {6, 7}, {0: 'zero', 1: 'one'}]
```

[ ]:
```
nested_list = [[1, 2], [3, 4], [5, 6]]

print(nested_list)
print(nested_list[0])
print(nested_list[0][1])
print(nested_list[1][1])
nested_list[1][1]=100
print(nested_list[1][1])
```

[ ]:
```
nested_set = {frozenset({1,2}), frozenset({3,4})}
print(nested_set )
```

[13]:
```
nested_dict = {
    "student1": {"name": "Anu", "age": 20},
    "student2": {"name": "Ravi", "age": 22}
}

print(nested_dict)
print(nested_dict["student1"])
print(nested_dict["student1"]["name"])
nested_dict["student1"]["name"] = "Mehann"
print(nested_dict["student1"]["name"])
```

```
{'student1': {'name': 'Anu', 'age': 20}, 'student2': {'name': 'Ravi', 'age': 22}}
{'name': 'Anu', 'age': 20}
Anu
Mehann
```

[19]:
```
t = ([1, 2], [3, 4])
print(t)
d = {"numbers": [1,2,3]}
print(d)
```

```
([1, 2], [3, 4])
{'numbers': [1, 2, 3]}
```

[20]:
```
t1 = (1, 2, 3)
list = [t1, (4,5)]
print(list)
set={(1,2,3),(3,4,5)}
print(set)
d = {(1,2): "A", (3,4): "B"}
print(d)

d = {"numbers": (1,2,3)}
print(d)
```

```
[(1, 2, 3), (4, 5)]
{(3, 4, 5), (1, 2, 3)}
{(1, 2): 'A', (3, 4): 'B'}
{'numbers': (1, 2, 3)}
```

[21]:
```
d1={'name':'Malar','usn':12345}
d2={'mark1':54,'mark2':87}
d1.update(d2)
print(d1)
```

```
{'name': 'Malar', 'usn': 12345, 'mark1': 54, 'mark2': 87}
```

[22]:
```
d1={'name':'Malar','usn':12345}
d2={'mark1':54,'mark2':87}
merge=d1|d2
print(merge)
```

```
{'name': 'Malar', 'usn': 12345, 'mark1': 54, 'mark2': 87}
```